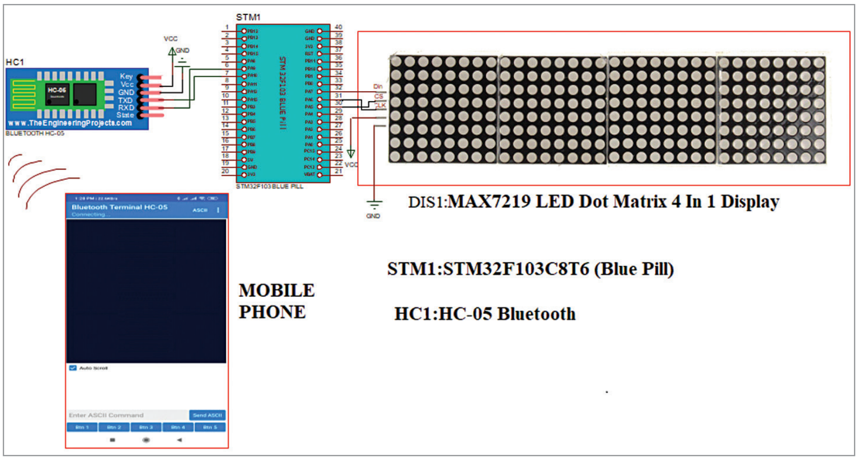

Fig. 6: Main circuit diagram

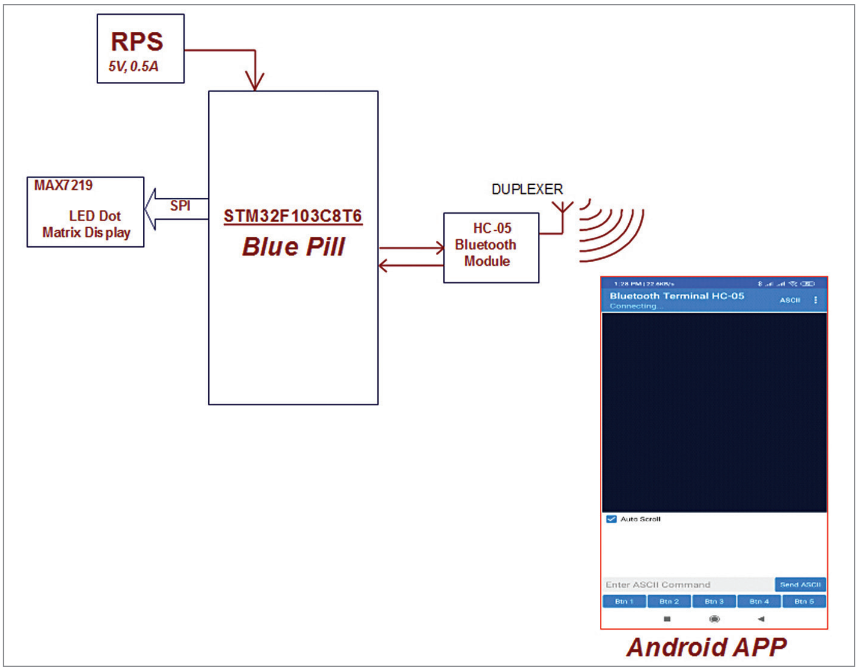

Fig. 7: Block diagram of project

IC1 is used to power the circuit.

The project requires 5V for the STM32 board, 3.3V ~ 6V for HC-05 Bluetooth module, and 3.3V ~ 5V for MAX7219 LED dot-matrix display.

The main control circuit, shown in Fig. 6, is built around STM32 with MAX7219 LED dot-matrix. The STM32 board is interfaced with an Android phone using Bluetooth. The 8 × 8 LED matrix connected to Arduino through MAX7219 is controlled through a dedicated application on the Android phone.

Since communication between STM32 and MAX7219 is based on SPI communication protocol, all we need is three pins from STM32 (Data, Clock, and Chip Select). The CS, CLK, and DIN pins of the MAX7219 IC board are connected to pins PA4, PA5, and PA7 of the STM32 controller.

Since Bluetooth is to be used for connection between STM32 and the Android device, the RX and TX pins of the HC-05 Bluetooth module are connected to TX1(PA9) and RX1(PA10) pins of the STM32 controller.

The block diagram in Fig. 7 shows interfacing of MAX7219 LED dot-matrix display with STM32 microcontroller using Arduino IDE and HC-05 Bluetooth.

Controlling the LED matrix through Android app

A dedicated app for Android based devices has been designed for this project. The layout of the app, which is already installed on a mobile phone (Bluetooth Terminal HC-05), is readily available in Google Play store from the free downloadable link *https://play.google.com/store/apps/details?id=project.bluetoothterminal&hl=en_IN*

Front view of the app is shown in Fig. 8.

The app contains two modes—the HEX format and the ASCII format. Here we have used the ASCII format mode. In this mode we can send text, numbers, and any special characters to display, by default.

The moment we type these in the terminal app, the STM32 microcontroller immediately receives the information serially through Bluetooth. The received information is segregated and the text is shown on the LED dot-matrix misplay through SPI protocol. Remember, the end characters of text \r \n must be Enable in the ASCII mode, as shown in Fig. 9.

(***EFY Note.*** The app utilises Bluetooth feature of the phone. Hence, necessary permissions must be given. Also, the HC-05 Bluetooth module must be paired with the device (phone), for which the default password is 1234.)

Software

The STM32 is just another microcontroller from STMicroelectronics. So, all the existing methods to program an ARM chip can be used for the STM32 board as well. One famous and commonly used IDE is the Keil ARM MDK, but we can also use IAR Workbench, Atollic TrueStudio, MicroC Pro ARM, Crossworks ARM,